

Based on 49 sources

Structure				
Name	Algebraic Properties	Cognitive Role	Physical Interpretation	References
Ξ-Δ Manifold	Filtered chain complex; non-commutative attention observables ($[A_i, A_j] \neq 0$); emergent geometric limit of octonionic non-associativity.	Models intelligence as a recursive process acting on filtration layers; uses Socratic Engine for gap detection; topologically invariant closure.	Coupled to spacetime torsion and prime singularities; macroscopic limit of microscopic algebraic energy; physical curvature ($P_n \cong P_{drift}$).	[1], [2]
Everything Equation Identity ($L = \Omega \Delta \partial(L)$)	Fixed-point identity of law object L ; composite of boundary involution ∂ , collapse Δ , and reflective closure Ω .	Recursive selection of lawful structure; Ω -aligned cognitive fields; safe recursive cognition via law-bound attractors.	Found in renormalization group universality, Hamiltonian constraint evolution, Maxwell's equations, and Einstein field equations.	Theorem 11.6, Definition 12.8, Section 10.6
Consciousness Tensor ($C_{\mu\nu}$)	Rank-2 tensor measuring local, windowed covariance between observables and their time-updates.	Measures a system's proprioception and the degree to which it tracks its own dynamics.	A physically real structure of self-reference; couples to physical observables like stress-energy through minimal interactions.	Section 9.1
Grothendieck	Category equivalent to	Conceptual world	Generalization of	Proposition